

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
6 FT 1 HT	(a)	(i)		Both springs stretch elastically (over the range) (1) Spring 2 stretches easier than spring 1 / up to 20 N spring 1 extends less / after 20 N /spring 1 extends more (1)			2	2	1	2
		(ii)		Pair of values from graph e.g. 47.5/10 (2) Answer = 4.75/4.8 N/cm (1) Allow tolerance of one small square when reading values from graph.		3		3	3	3
		(iii)		Spring 1- all points on the line OR converse for spring 2		1		1		1
	(b)	(i)		Scales – x axis: each large square = 4/5/6 cm; y axis: 1 large square = 10 N (1) 7 points correct $\pm$ < one small square tolerance (2) 6 points correct $\pm$ < one small square tolerance (1) 5 or less (0) straight line until 50 N then curve $\pm$ < a square tolerance (1)		4		4	4	4
		(ii)	I	E labelled where line starts to curve (e.g.12, 50)		1		1		1
			II	e.g. 50 N (follows from prior answer)		1		1		1
		(iii)		Extension is no longer proportional to extension (when it is 30 cm) / past elastic limit (1) So spring will not return to its original length so not suitable (1)			2	2		2
				Question 6/1 total 14		10	4	14	8	14